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# Improving instructional design proficiency of master's students in mathematics education through intelligent educational technologies

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ABSTRACT

Research on smart education has predominantly focused on macro-level concepts and frameworks, such as core principles and system architectures, while offering limited guidance for discipline-specific integration and practical implementation. This study addresses two critical gaps: (1) how an M.Ed. course can be enhanced through intelligent educational technologies, and (2) whether a smart classroom teaching mode can effectively improve M.Ed. students' instructional design competence, particularly in terms of precision and professionalism.

Adopting a pedagogically oriented approach rather than an algorithmic one, this study proposes a three-dimensional framework encompassing learning effectiveness, information and communication technology (ICT), and classroom organization. Based on this framework, a set of smart classroom characteristics aligned with M.Ed. program objectives is developed. Using a mathematics M.Ed. course as a case study, the study integrates intelligent educational technologies—such as automated scoring, personalized recommendations, and multi-AI feedback—across the pre-class, in-class, and post-class stages to construct a technology-enhanced teaching loop.

Results from a quasi-experimental design and quantitative analysis indicate that this approach significantly improves students' ability to formulate precise and professionally grounded instructional objectives. Building on these findings, the study proposes the "D–T–E Model" (Disciplinary Demand–Technological Empowerment–Evaluation Loop) as a transferable pedagogical framework for professional master's education. The findings provide practical implications for the design and implementation of AI-supported smart classrooms in teacher education contexts.

## 1. Introduction

The rapid development of smart campuses is driving the digital transformation of higher education, with increasing integration of intelligent technologies into teaching and learning environments. In particular, intelligent learning systems have been progressively adopted in classroom settings (Díaz-Parra et al., 2022).However, traditional instructional models remain constrained by standardized resources, limited spatiotemporal flexibility, and insufficient adaptability to diverse learner needs (Demir, 2021; Göksu & Atici, 2013; Tomlinson & Kalbfleisch, 1998; Zhu et al., 2016). These limitations have become more pronounced in the context of smart education, which emphasizes personalization, flexibility, and technology integration.

Current research on smart education has primarily focused on conceptual frameworks, curriculum design, and the construction of smart learning environments. However, it provides limited guidance for practical and discipline-specific implementation (Martín, A. C., Alario-Hoyos, C., & Kloos, C. D., 2019). A unified smart teaching mode may not always be compatible with all disciplines in higher education due to their diverse educational needs and varying implementation strategies. Specific research in a variety of specialties is necessary to effectively implement smart education.

Professional Master of Education (M.Ed.) programs in the social sciences have been extensively established worldwide. These programs aim to cultivate highly qualified, interdisciplinary professionals equipped for teaching and leadership roles in basic education (Yang, 2017). However, existing research has primarily focused on macro-level program objectives and structural models, with limited attention to course-level pedagogical practices. It also overlooks the rapid advancement of educational technologies and their role in shaping contemporary teaching practices. As a result, these technologies remain insufficiently integrated into M.Ed. pedagogy and curriculum design.

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As education undergoes digital and intelligent transformation, the training objectives for M.Ed. students are being redefined to reflect emerging demands. Traditional teaching models are increasingly inadequate for developing core competencies and addressing complex instructional contexts. This has led to a growing misalignment with current educational demands. The traditional postgraduate training model—typically supervisor-centered, knowledge-transmission-focused, and thesis-oriented—is under significant pressure to transform in light of rapid technological advances, particularly in artificial intelligence (Deng et al., 2025). Furthermore, a dominant "one-size-fits-all" approach continues to characterize higher education. It is largely defined by the unidirectional transmission of knowledge from instructor to student. First, it inadequately supports the reciprocal nature of teaching and learning, where both instructors and students grow through the process. Second, it offers limited practical mechanisms for addressing diverse learner profiles and implementing individualized instruction. Consequently, it fails to fully address the evolving demands of preparing M.Ed. graduates for contemporary educational challenges (Zhang & Ma, 2024).

This study examines a discipline-specific smart classroom framework for Master of Education (M.Ed.) programs, using mathematics instructional design as a case study. It integrates multi-AI feedback into the development of instructional design competencies and evaluates its impact on the precision and professionalism of instructional objective formulation. The findings provide empirical evidence for integrating intelligent technologies into subject-specific pedagogy. Rather than proposing a new smart education model, this study synthesizes existing technological tools to improve instructional design proficiency. It further develops a discipline-specific implementation framework for smart classroom instruction. By integrating the three dimensions of Learning Effectiveness, Information and Communication Technology (ICT), and Classroom Organization, this study explores and validates a practical implementation pathway. The framework offers a structured instructional design model for M.Ed. programs. It also serves as a transferable template that can be adapted to support the digital transformation of pedagogy in other disciplines.

The remainder of this study is organized as follows. Section 2 reviews the literature on smart classrooms and the development of M.Ed. programs. Section 3 outlines the research methodology. Section 4 describes the system of smart class characteristics and teaching modes for M.Ed. students. Section 5 reports and discusses the findings, and Section 6 concludes the study with implications and limitations.

## 2. Literature review

### 2.1. Smart education

#### 2.1.1. Definition of smart education

The integration of education and information technology has developed rapidly over the past two decades. Early efforts emphasized the use of multimedia and network technologies to enhance pedagogical methods and improve instructional efficiency. Recent advances in Artificial Intelligence, big data, and mobile technologies have shifted the focus toward smart education. This paradigm emphasizes adaptive, personalized, and data-driven learning environments.

Although the term "smart education" is widely used, a unified definition within academia remains elusive. Instead of a singular theory, smart education is generally understood as an educational paradigm defined by a set of core characteristics. Zhu et al. (2016) established a foundational research framework that outlines key components for fostering "smart learners". Their framework integrates a four-tier structure of smart pedagogy (including differentiated, collaborative, personalized, and generative learning) along with key features of smart learning environments and supporting technological infrastructures. Crucially, this work emphasizes that the "smartness" of education does not arise from technology alone but emerges from the synergy between pedagogical design, technological context, and learner development. It thus provides a crucial integrative perspective for understanding subsequent interpretations of smart education. Building upon such holistic perspectives, researchers have since approached and conceptualized smart education from various, more specific perspectives.

From a technology application perspective, smart education is often defined in terms of technological infrastructure and data use. For instance, Wang et al. (2020) argue that data mining and big data-driven learning analytics constitute the core support for smart education because they significantly enhance the assessment of student learning and support the refinement of pedagogical practices. Under such definitions,the"smart "in smart education refers to the technological apparatus rather than the educational process or outcomes. Similarly, Wang et al. (2020) emphasize that the smart educational technologies should be able to identify and respond to the needs and conditions of various participants. Taken together, these definitions suggest that the "smartness" of smart education is primarily attributed to the sophistication of the technology itself rather than to the educational processes or outcomes it enables. From this perspective, technology is fundamentally a means to an end, not the end goal of education.

From the perspective of defining characteristics, Yun et al. (2017) identify five key attributes of smart education: ICT (T), open learning environment (E), self-directed learning (S), tailor-made learning (C), and social learning (SL). Similarly, Glukhov and Vasetskaya (2 017) contend that the core feature of smart education is the provision of personalized learning pathways that accommodate student preferences and abilities, and support their development.

From the perspective of educational value, Javaid, Haleem, Singh, and Suman (2022) conceptualize smart education as a reconstruction of educational processes aimed at cultivating advanced thinking and creativity through integrated smart teaching, learning, evaluation, and management.

Synthesizing these perspectives, smart education can be understood as a learner-centric paradigm that uses advanced technologies to create open, flexible, and personalized educational experiences. Its ultimate goal is to develop adaptable and competent learners. Tikhomirov et al. (2015) argue that the primary aspect of smart education is not the ICT component, which functions primarily as a tool for achieving specific educational outcomes. Considering dimensions beyond technology, they established a paradigm for smart education, describing its form through three dimensions: educational outcomes, ICT, and organization. Accordingly, this study adopts and adapts this three-dimensional research paradigm proposed by Tikhomirov et al. (2015) as its core analytical framework. Other key elements emphasized by scholars, such as personalization, environmental openness, and self-direction, are operationalized as specific indicators within these three core dimensions.

#### 2.1.2. Smart classes in higher education

With the advancement of infrastructure in higher education, smart education has been increasingly implemented in universities.As a key implementation form of smart education, smart classrooms have been widely adopted in higher education.For example, an engineering institution in the Middle East implemented a technology-enhanced active learning classroom (ALC) design.Personal response systems, smart TVs, and ceiling projectors provide classroom flexibility for content delivery and interaction (Gunn & Raven, 2017). Similarly, King Saud University adopted a learning management system (LMS) aligned with its long-term development strategy to support the management, tracking, and evaluation of e-learning programs (Alelaiwi et al., 2015). In Indian universities, smart education has been applied in remote laboratories to support the training of engineering students (Gowripeddi et al., 2019). Gifu College, National Institute of Technology in Japan, has enhanced engineering education by integrating ICT infrastructure, curriculum design, and digital learning resources (Ogawa & Shimizu, 2016).In higher education, smart classroom models have been widely applied to support learning in science and engineering disciplines. These applications include real-time lecture broadcasting, the use of augmented and virtual reality for experimental learning, and visualizations to support conceptual understanding (Alelaiwi et al., 2015; Li et al., 2021).

### 2.2. Cultivation for the Master of Education

To meet the demand for qualified professionals in basic education, China established the in-service Master of Education (M.Ed.) program in 1996 and later introduced a full-time M.Ed. program in 2009. The M.Ed. is a professional degree designed to develop practitioners with a solid theoretical foundation, research competence, and practical teaching expertise.

The expansion of M.Ed. programs in China reflects evolving societal demands for educational professionals (Jung, 2020). Sui (2017) highlights the growing need for professionally oriented M.Ed. programs in response to these changes. The universalization of higher education and the transformation of China's economic and industrial structure have created new demands for high-quality educational professionals. This shift requires M.Ed. programs to better align with both students' aspirations for advanced education and the needs of a rapidly evolving labor market. Liu (2014) argues that pre-service teacher education in China tends to prioritize technical skills over reflective and research-oriented competencies, resulting in limited innovation and potential constraints on educational development. Collectively, these studies underscore that, within the context of modern high-quality development, the primary objective of M.Ed. programs is to elevate the academic qualifications and professional quality of teachers, specifically by enhancing their research capabilities and praxis-oriented competencies.

Most universities in China follow the U-S collaborative mode of training for undergraduate teacher trainees, with the joint training of university (U) and primary and secondary school (S). However, unlike undergraduate teacher training, M.Ed. programs aim to cultivate students who possess research abilities and high-level professional research competence. It is difficult, however, to enhance the professional research ability of the M.Ed. via the U-S collaborative mode. Regarding the innovation of the training model of M.Ed., the two prominent models are the U-T-S and G-U-S joint training modes. Allowing basic education research and development institutions (T) or local governments (G) to act as intermediaries between universities and primary or secondary schools can change the problems that need to be solved to cultivate an M.Ed. and help cultivate expert clinical teachers (Li & Sun, 2022; Zuo et al., 2023).

Wang (2022) identified the Community, Immersion, and Four-Stage Interactive Modes based on the lack of cooperation, insufficient depth, and insufficient connection in the practice teaching of M.Ed. Ma and Zhao (2022) developed a four-in-one teaching mode of doing, researching, editing, and learning based on original cases, which helps M.Ed. graduate students establish a substantial connection between the closed and logical theoretical system and the open and complex educational practice, enabling close interaction and cross-promotion between students' learning, practicing, and research processes. Sun (2020) suggested a three-dimensional teaching mode of specialization, and research for the Master of Music Education. Xiong et al. (2020) proposed modular teaching, micro-class-assisted teaching, and open teaching modes for Master of Biology Education. Qi et al. (2020) proposed open access, digitalization, and virtual simulation technology combined with a traditional experimental teaching mode for a Master of Physics Education.

### 2.3. Current research status of smart education in the field of Master of Education development

Current research on smart education has predominantly operated at the macro level, focusing on conceptual elaborations of smart classrooms, curriculum frameworks, and instructional design models (Kim & Bae, 2013). However, there remains a relative lack of pedagogical models tailored to specific disciplinary contexts. Existing empirical studies are largely concentrated in engineering education, while applications within social science domains remain underexplored. This imbalance is particularly problematic given that smart education offers affordances—such as personalized learning guidance, data-driven analytics, open resource ecosystems, and high-frequency interactive environments—that are highly relevant for education-related disciplines.These technological supports are equally, if not more, essential for professional programs in education.The advancement of modern educational technology serves as a pivotal contextual backdrop; as student needs evolve, educational systems must adapt to become more intelligent.

The technologization and modernization of education constitute an ongoing structural transformation, within which both educational technology and the professional training of Master of Education (M.Ed.) students serve as critical contextual foundations (Singh & Miah, 2020). Nevertheless, prior research on smart classroom applications in M.Ed. programs has been predominantly discipline-specific (e.g., physics; Qi et al., 2020) and tends to emphasize technological integration rather than the construction of coherent smart pedagogical models. In the case of mathematics education at the postgraduate level—particularly within social science-oriented higher education contexts—systematic exploration of smart teaching modes remains limited.

Accordingly, several key research questions emerge: How can mathematics instruction in M.Ed. programs be reconceptualized through the lens of smart education? What pedagogical models are most appropriate for this context? And what measurable effects do such models produce? Addressing these questions is essential for advancing both the theoretical development and practical implementation of smart education in discipline-specific contexts.

## 3. Methodology

This study adopts a two-phase research design consisting of an exploratory qualitative phase and a quasi-experimental phase. Phase 1 focuses on identifying the systemic characteristics of smart classrooms in Master of Education (M.Ed.) programs and develops a discipline-specific teaching model, with mathematics education serving as the focal context. Phase 2 examines the impact of the proposed teaching model on students' learning outcomes.

The first phase employed a qualitative approach to identify and conceptualize key features of smart classrooms. Guided by smart education theory, the study identified a set of core characteristics aligned with the pedagogical goals of M.Ed. programs. These characteristics informed the development of a smart classroom teaching model, which is detailed in Section 4 along with the full research process and findings.

The second phase adopted a quasi-experimental design with pretest–posttest control group structure. Participants were graduate students enrolled in a mathematics education M.Ed. program, and were assigned to either an experimental or control group. The experimental group received instruction based on the proposed smart classroom teaching model, focusing on the design of instructional objectives for high school mathematics. The control group received conventional instruction on the same topic. Data were collected through pre- and post-intervention assessments. Descriptive statistics and inferential analyses were conducted using SPSS Version 26 to compare learning outcomes between groups.

The participants in this teaching experiment were 68 master students enrolled in the Mathematics Education track of the Subject Teaching program at the College of Science, Jimei University. The course was delivered by a professor with substantial teaching experience at the university. Prior to the formal experiment, all 68 participants completed a pretest and were then assigned to either an experimental or a control group using a matched-pair procedure to ensure baseline equivalence. During group assignment, care was taken to ensure that the difference in mean scores between the two groups was within 0.5 points. Prior to conducting the independent-samples *t*-test, we verified the assumptions of normality and homogeneity of variance. The Shapiro–Wilk test was used to assess normality. Results showed no significant deviation from normality for either the control group ($W = 0.976$, $p = 0.612$) or the experimental group ($W = 0.983$, $p = 0.847$). Levene's test was employed to examine homogeneity of variance, with no significant difference between group variances ($F = 0.032$, $p = 0.859$). These results supported the use of an independent-samples *t*-test for subsequent analyses. An independent-samples *t*-test was conducted to compare the pre-test scores of the two groups. The results showed no statistically significant difference between the groups ($t = 0.124$, $p = 0.902$), confirming baseline equivalence. To enhance the contextual relevance of the evaluation rubric, two curriculum experts and three experienced secondary mathematics teachers were invited to refine its wording. They then independently scored all 68 instructional design assignments. Inter-rater reliability was assessed using the intraclass correlation coefficient (ICC). The results showed an average measure ICC of 0.85 (95% CI: [0.78, 0.90]), indicating a high degree of consistency among the experts' ratings and confirming the reliability of the scoring data.

This study quantitatively compared the instructional objective design scores of students in the experimental and control groups to assess the impact of the smart classroom intervention. The efficacy of the smart classroom approach was examined by comparing learning outcomes between the smart classroom and traditional instruction conditions. The study used Zhang's (2021) instructional objective design rubric as the evaluation instrument, which was adapted for the M.Ed. context by revising specific descriptors to enhance contextual relevance (see Table A.1 in the Appendix). The adapted rubric achieved a Scale-level Content Validity Index (S-CVI) of 0.92, indicating good content validity in this new application context. The rubric's operationalization of instructional design competency is theoretically grounded in foundational frameworks from the field of instructional design. For example, Mager's (1962) behavioral objectives theory—emphasizing that objectives must be observable and measurable—informs two rubric indicators: T1 (Standard Compliance), which requires the use of measurable action verbs, and T5 (Knowledge Essence), which assesses whether the essential attributes and logical structure of the content are clearly articulated. Two additional theoretical frameworks underpin the rubric. First, Shulman's (1987) construct of pedagogical content knowledge (PCK)—defined as the specialized capacity to transform subject matter into forms accessible to learners—was operationalized through T2 (Competency Development), which links core competencies to specific mathematical content, and T7–T9 (Learner Analysis), which evaluates alignment with students' prior knowledge, cognitive abilities, and experiential backgrounds. Second, Biggs' (2011) principle of constructive alignment, which posits that learning objectives, teaching activities, and assessment must form a coherent system, informed T3 (Holistic Integration), assessing whether the instructional design establishes a hierarchically organized content structure.

Data analysis was conducted using IBM SPSS Statistics (Version 26). Descriptive statistics were computed for both groups ($n = 34$ per group), including means for: (a) the overall instructional objective design score; (b) each primary indicator; and (c) all sub-indicators nested within each primary indicator. An independent-samples *t*-test was conducted to compare instructional objective design scores between the two groups. The results are reported in Section 5.

## 4. Constructing the smart class model for Master of Education cultivation

### 4.1. Smart class characteristics system for Master of Education cultivation

This study aims to construct an analytical framework for smart classrooms tailored to the training of Master of Education (M.Ed.) students. To achieve this goal, an integrated three-dimensional analytical framework has been developed and adopted. The framework is theoretically grounded in the smart education paradigm proposed by Tikhomirov et al. (2015), which conceptualizes smart education as a system driven by the synergy of "educational outcomes, technology, and organization." We further integrate this paradigm with the core tenets of outcome-based education theory (Biggs & Tang, 2011) and technology integration research (e.g., Zhu et al., 2016). Through this theoretical integration, three analytically distinct yet interrelated dimensions are constructed: learning effectiveness, information and communication technology (ICT), and classroom organization.

These three dimensions together constitute an integrated analytical framework, with each dimension addressing a distinct aspect of the smart classroom: The learning effectiveness dimension addresses the core question of "what the smart classroom ultimately aims to achieve." It is operationalized as assessable and observable professional competencies that M.Ed. students are expected to acquire This dimension provides a reference for aligning technological applications and pedagogical organization with explicit educational goals. The information and communication technology (ICT) dimension, grounded in research on technology integration, examines how specific technologies (e.g., learning analytics and AI tools) are designed and implemented within instructional contexts. It focuses on their roles in enabling personalized learning pathways, supporting data-informed instructional decisions, and providing timely feedback. In this way, the ICT dimension functions as the enabling mechanism for achieving the intended learning outcomes. The classroom organization dimension focuses on the instructional context in which smart classroom practices are enacted. It examines how teaching interactions, learning environments, and teacher–student roles are reorganized in response to smart classroom implementation.This dimension highlights the conditions under which technology integration and pedagogical innovation can be meaningfully embedded in authentic M.Ed. teaching contexts.

Building on the proposed framework, the study operationalizes it within the context of M.Ed. training by examining how each dimension is instantiated in teacher education practice. This process results in the identification of a set of domain-specific features for the M.Ed. smart classroom, as presented in Fig. 1.

Learning effectiveness refers to the intended and achieved learning outcomes of instruction, typically manifested in students' knowledge acquisition, cognitive skills, and professional competencies. Different dimensions of learning effectiveness are influenced by corresponding instructional structures, learning environments, and pedagogical designs. Within the content of M.Ed. training, this study identifies the dimensions of learning effectiveness as knowledge content, research abilities, and practical abilities.

Whether in traditional or modern smart teaching environments, knowledge content remains the cornerstone of classroom learning and the vehicle for the transmission of thought. Consequently, knowledge acquisition stands as a basic objective and a significant outcome of instruction. The knowledge content for M.Ed. students is characterized by its interdisciplinary nature. Taking Subject Mathematics as an example, M.Ed. classroom instruction must balance broad educational theories with the imparting of specialized mathematical knowledge. Instructional design requires a degree of integration and fusion of these elements to form a cohesive knowledge system.

The inclusion of research and practical abilities is grounded in the training objectives of M.Ed. programs in China. Research ability is conceptualized as comprising three components: problem identification, research competence, and innovative thinking. The process of moving from problem identification to problem solving constitutes a fundamental trajectory of academic inquiry and represents a core competence expected of M.Ed. students. Innovative thinking is the cognitive foundation for discovering and solving research problems effectively.

Practical ability is further conceptualized as consisting of three components: foundational teaching skills, professional dispositions, and

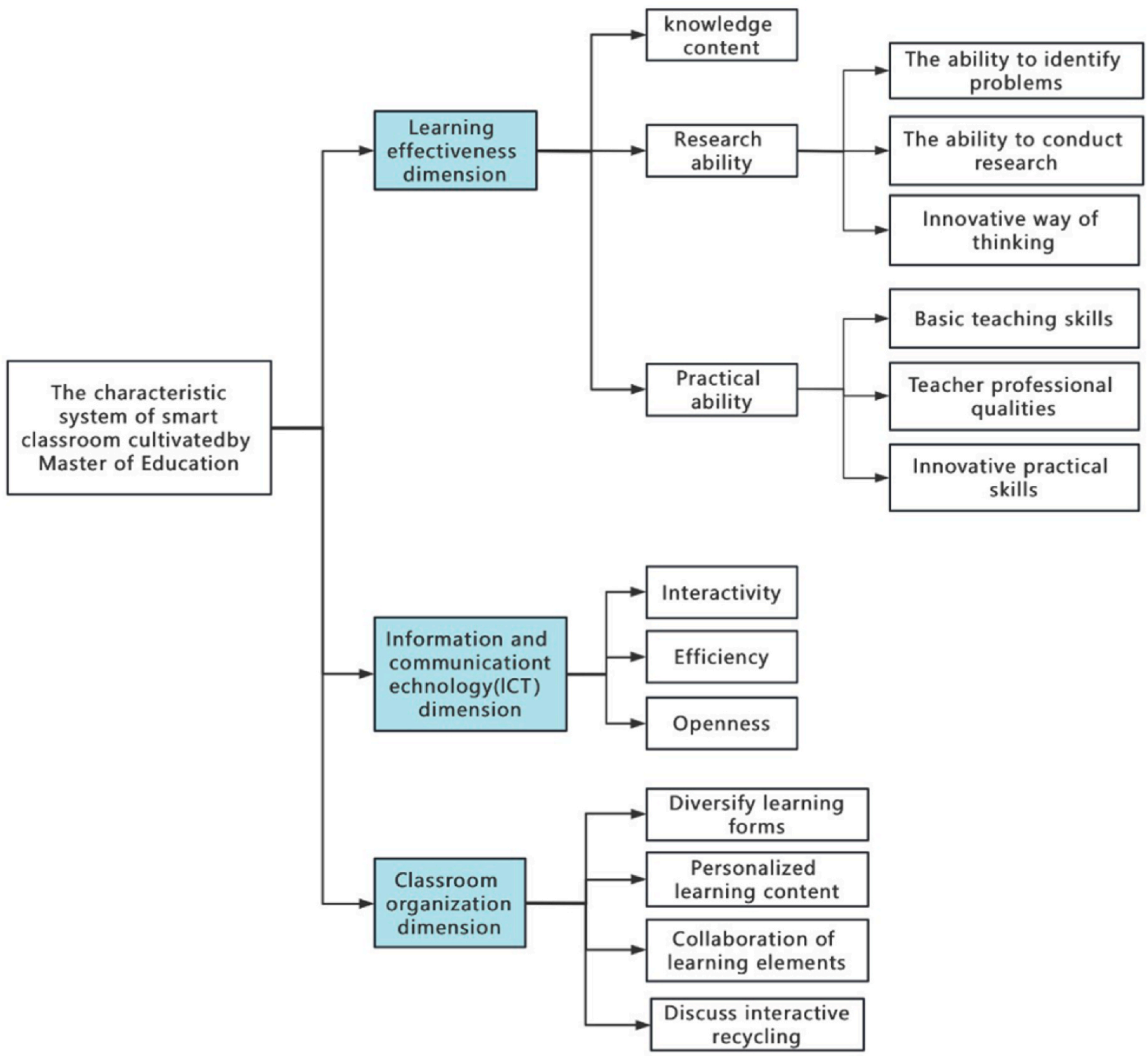


**Fig. 1.** System of Smart Classroom characteristics for M.Ed. Programs (Original flow chart).

innovative practice. Foundational teaching skills and professional dispositions are essential for M.Ed. graduates to effectively enter and function within teaching roles. Innovative practice refers to the ability to translate pedagogical ideas into practice. This is a requirement for the M. Ed. For their professional development.

The ICT dimension comprises a suite of tools designed to facilitate various aspects of smart education. This suite includes three distinct categories.

1. Management Tools: The first category encompasses tools for organizing and managing educational processes, providing big data processing capabilities regarding these processes and enabling automated management.
2. Content Development Tools: The second category consists of specialized software designed for developing educational content. This may include tools for creating text and multimedia materials for curriculum development. Crucially, tools for developing smart learning content must offer high interactivity and create effects of virtual presence or telepresence.
3. Social Interaction Tools: The third category is based on social interaction platforms, particularly social networks and webinar software, which allow students to communicate with instructors and peers.

Unlike engineering disciplines that require virtual laboratory environments, the smart classroom for M.Ed. training does not necessitate such simulations. Virtual environments are neither essential nor necessarily effective for M.Ed. classroom instruction. Educational learning outcomes are characterized by their indeterminate nature; the learning process involves the sharing of educational perspectives and the interaction of thoughts on educational issues between teachers and students. Yu et al. (2022) demonstrated that, compared to traditional classrooms, smart classroom instruction promotes student engagement. Accordingly, this study identifies the characteristics of the M.Ed. smart classroom within the ICT dimension as: Interactivity, Efficiency, and Openness.

The classroom organization dimension of smart education is characterized by flexibility, diverse instructional formats, openness, knowledge management practices, and personalization. These characteristics support the design of adaptive learning pathways that accommodate individual student needs. This process should incorporate diverse forms of learning, appropriate ICT infrastructure, innovative educational technologies, and pedagogical design to ensure freedom in student learning. Based on this framework, the classroom organization dimension of the M.Ed. smart classroom is characterized by five key features: diverse learning formats, personalized learning content, collaborative learning structures, diversified assessment strategies, and iterative cycles of discussion and interaction.

The smart class can support students' independent learning, assist students in conducting mutual studies, and aid teachers in tracking students' progress via functions not limited to lectures. Smart classes personalize teaching by relying on the teaching platform, cloud computing services, and artificial intelligence technology. Smart learning platforms can deliver personalized and differentiated learning resources, enabling students to access and utilize information more effectively. A smart class integrates learning interactions and resources with the assistance of a smart teaching system. This process facilitates collaboration between learning elements. Assessment in smart classrooms is multifaceted. It includes teacher assessment, platform-based analytics, and peer or self-assessment. Smart learning platforms can record interaction and discussion processes across learning stages. Teacher–student interaction can extend beyond the classroom, forming iterative cycles of discussion and reflection. This multi-stage, iterative interaction contributes significantly to the enhancement of students' cognitive levels.

Drawing on the proposed framework, this study develops a classroom teaching model structured around Disciplinary Demand, Technology Empowerment, and an Evaluation Loop, hereafter referred to as the D–T–E model.

### 4.2. *Smart class teaching model for Master of Education cultivation*

Based on the smart classroom framework for Master of Education programs, this section utilizes *a high school mathematics lesson on Instructional Objective Design* as a case study to demonstrate the "D-T-E Model." As illustrated in Fig. 2, the model's instructional flow is guided by specific.smart classroom characteristics across three dimensions: educational outcomes, ICT, and organization. These characteristics are systematically integrated into each phase of the teaching process.

#### 4.2.1. *Activities before class*

The "D-T-E Model" leverages the smart teaching platform to precisely deliver "specialized practical tasks" and "pre-class guided exercises" focused on high school mathematics instructional objective design. Upon completing their designs independently, students upload their

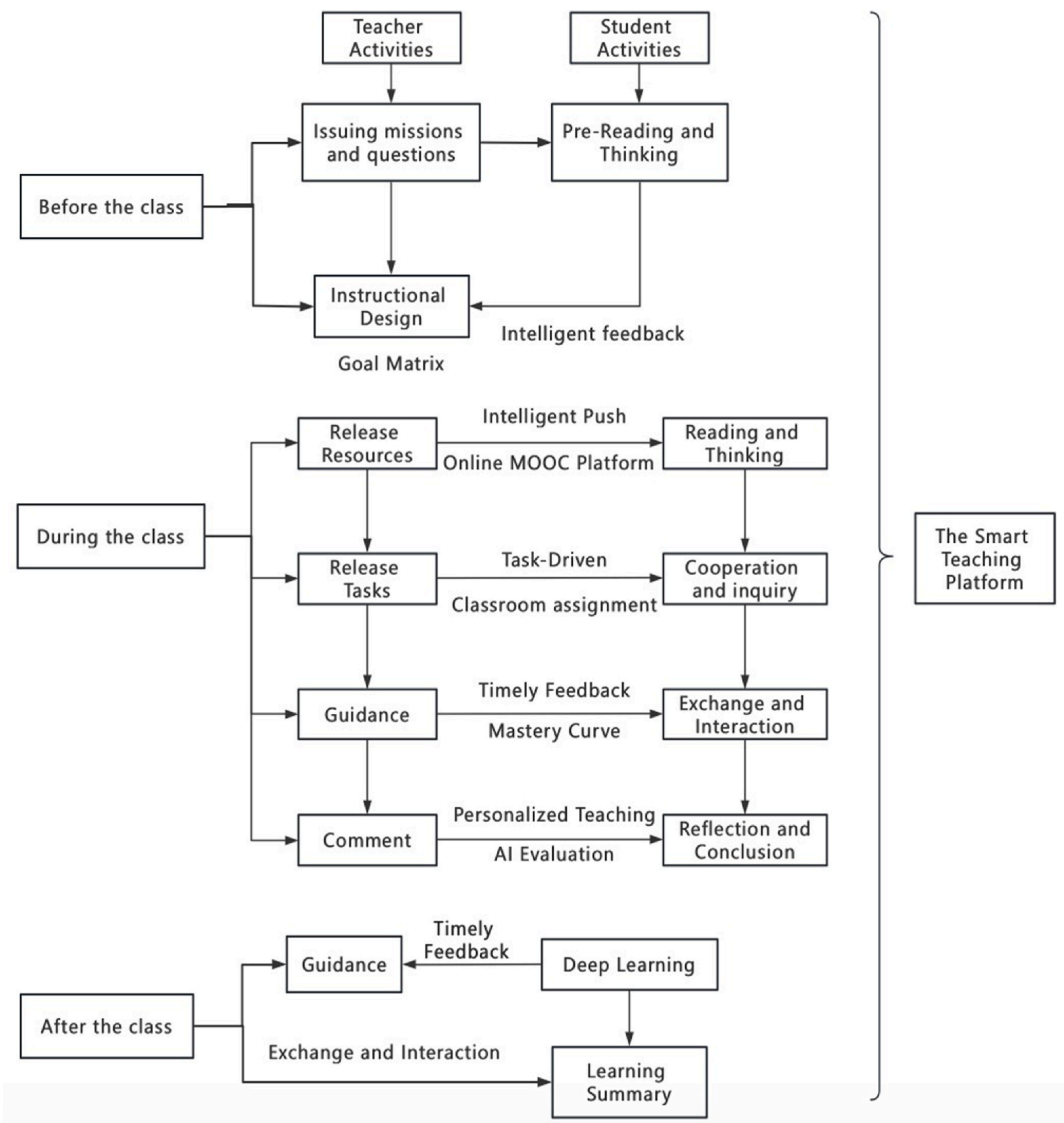


**Fig. 2.** Smart Classroom Teaching Process for the M.Ed. Program (Original flow chart).

outcomes to the platform in text or image format. Utilizing a preset evaluation rubric (see Table A.2 in the Appendix) and a parallel computing system, the platform executes rapid intelligent scoring and multidimensional data analysis to generate visualized evaluation results. This process significantly reduces the time latency associated with traditional manual assessment. Consequently, it provides robust data support for teachers to accurately grasp learning conditions and conduct targeted instruction, thereby facilitating personalized and highly efficient classroom teaching.

During the pre-class exercise phase, the smart platform automatically logs key data such as students' response time and accuracy rates, and visualizes the data through intuitive charts (e.g., pie and bar charts). Concurrently, student difficulties are categorized into five distinct types: (1) Ambiguity in task comprehension: Unclear understanding of the specific requirements and scoring dimensions of *the Instructional Objective Design Evaluation Rubric.* (2) Difficulty in textbook content translation: Challenges in transforming textbook content into measurable and assessable instructional objectives. (3)Information overload: Difficulty in discerning valid information amidst massive resources and conflicting viewpoints. (4)Superficial cognitive processing:Thinking remains at a surface level, lacking deep analysis of the textbook's editorial logic and key difficulties. (5)Technical inefficiency:Unfamiliarity with platform operations Based on the platform's intelligent feedback report, instructors can precisely assess students'cognitive levels regarding instructional objectives, their mastery of textbook content. This allows for the identification of knowledge gaps, enabling teachers to scientifically optimize the focus of in-class instruction and design targeted collaborative inquiry tasks in advance (see Fig. 3).

On this basis, teachers can implement stratified grouping based on student differences and map out personalized learning paths. For students struggling with knowledge application: The system recommends videos interpreting the High School Mathematics Curriculum Standards and exemplar sets demonstrating the transformation of "textbook content into instructional objectives." This scaffolding assists students in grasping fundamental theories and achieving effective translation. For students facing difficulties in information selection: The system provides curated packages of authoritative literature on instructional design and navigational maps of high-quality resources. This guidance facilitates the efficient screening and integration of information.

Concurrently, instructors publish inquiry-driven core questions on the platform to guide student exploration. Examples of such questions include: "What are the theoretical underpinnings of your instructional objective design?""What innovative insights have emerged from your analysis of the textbook?""What is the underlying logic of objective setting within the Curriculum Standards?" "Do you hold alternative perspectives that warrant critical discussion?" Guided by these inquiries, students utilize the campus network to access domestic and international academic databases and MOOCs. By independently studying relevant pedagogical theories and absorbing cutting-edge viewpoints, students not only expand their learning boundaries, but also cultivate their research capabilities and capacity for practical innovation.

### 4.2.2. *Activities during the class*

The "D-T-E Model" aggregates and uploads curated instructional resources to the smart platform, including expert interpretations of curriculum standards and key competencies, as well as exemplary cases of instructional design. To address individual differences, the platform employs a personalized intelligent push mechanism. Driven by data analytics, this mechanism matches learning resources to each student's specific cognitive level, performance history, and affective attributes. Complementing this automated delivery, students retain the autonomy to actively retrieve resources from the teacher-curated repository based on their specific needs and preferences.

Following the review of intelligently pushed resources, the formal instructional session commences. After the initial exposition of the content, the teacher facilitates a discussion centered on specific classroom inquiries. During this discourse, the teacher iteratively refines and augments the questions to deepen student engagement. This process realizes the recirculation of communication and interaction.

The specific class activities are detailed in Table A.3(see the Appendix). Upon completion of these activities, students will refine their instructional objective designs by synthesizing the insights gained from the discussions.

### 4.2.3. *Activities after the class*

Continuous and timely evaluation is critical to enhancing learning quality (Tian,x.c., Deng,Y.c.& Zhang,T.2020). To enhance the instructional objective design competence of Master of Education (M.Ed.) students, the "D-T-E Model" integrates Artificial Intelligence as a mechanism for immediate feedback. Upon completing their initial drafts, students submit their work to three mainstream Large Language Models (LLMs)—DeepSeek, Doubao, and Kimi. Acting as "intelligent reviewers," these models diagnose the designs across four dimensions: scientific rigor, adaptability, operability, and developmental potential, providing specific suggestions for revision.Subsequently, students synthesize the tripartite AI feedback to conduct self- and peer-assessments on the platform. Students revise their drafts based on AI suggestions, followed by instructor evaluation. This iterative cycle continues until

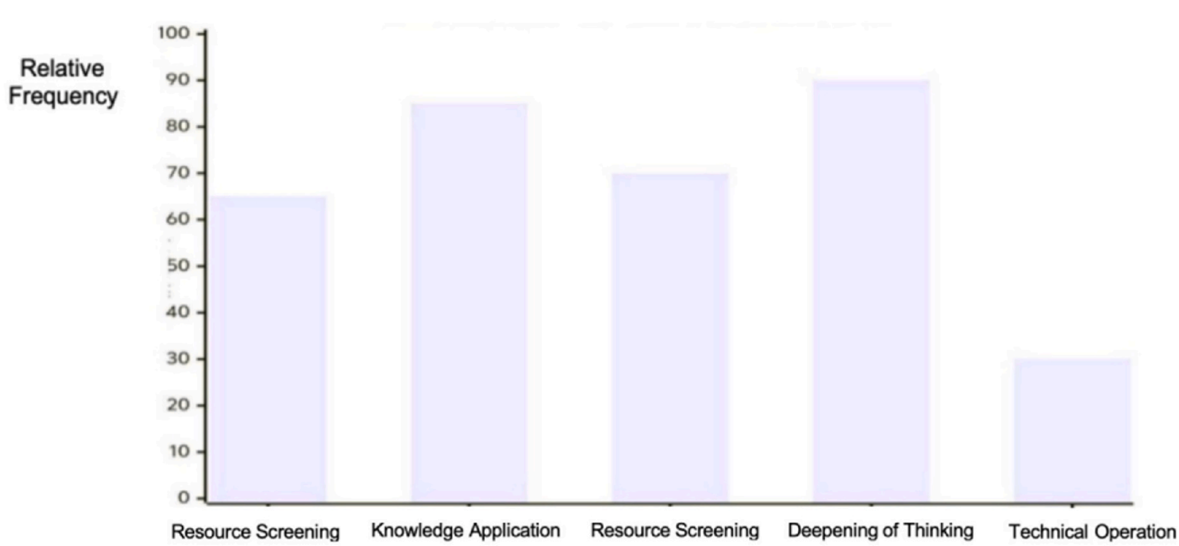


**Fig. 3.** Frequency analysis of problems encountered by students when completing pre-class tasks.

the final instructional objective design is completed and uploaded to the smart platform. In this way, the traditional single-submission assignment is transformed into a closed-loop learning cycle of "design--feedback-reflection-optimization," enabling students to progressively refine their design capabilities through sustained human-AI interaction.

The researchers utilized the Instructional Objective Design Quality Evaluation Rubric (see Fig. 4) to "train" the AI, ensuring it fully comprehended the criteria for judging the quality of instructional objectives. Subsequently, student submissions were input into the AI for evaluation. The following is the initial draft of instructional objectives written by Student S on the topic of "Monotonicity of Functions".

1. Be able to articulate the definitions of increasing and decreasing functions using mathematical language.
2. Be able to determine the monotonicity of functions through graphical and definitional methods.
3. Understand the monotonicity of functions through examples and familiar function graphs, and correctly use symbolic language to characterize it.
4. Enhance core competencies such as intuitive imagination and mathematical abstraction; appreciate the "conciseness" of mathematical symbols and learn to express real-life mathematics using symbolic language.

This draft was input into three of the most widely used Chinese Large Language Models (LLMs) for scoring feedback. The evaluation feedback obtained is as follows (see Table A.4 in the Appendix):

These models demonstrate robust instruction-following and text comprehension capabilities within the Chinese linguistic context, enabling the effective evaluation of the logic, standardization, and measurability of instructional design artifacts, such as instructional objectives. Furthermore, given that these three models are developed by different institutions, there are differences in training data and model architecture, they frequently yield complementary or even diverse evaluation opinions. This diversity exposes students to multiple perspectives during the revision process, thereby promoting critical reflection. This "triangular mutual verification" strategy via AI feedback aligns with the core concept of this study, which aims to enhance the ability to design instructional objectives through timely, diverse, and operable intelligent feedback.

Through comparison, we found that the three AI models each have distinct characteristics in supporting instructional objective design.

1. DeepSeek excels in theoretical construction, providing logically rigorous and structurally standardized analyses from a curriculum theory perspective.This capability helps learners understand the connotation of the three-dimensional objectives and is suitable for consolidating the academic foundation of instructional design.
2. Kimi focuses on instructional practice,it precisely diagnoses deficiencies in objectives and proposes practical strategies, thereby facilitating the rapid optimization of lesson plans.
3. Doubao is characterized by conciseness and efficiency, It provides clear, direct directions for modification, making it ideal for time-constrained scenarios.

In guiding M.Ed. students, these three models can be employed synergistically: Kimi to construct the knowledge and skill framework, Doubao to ensure objectives are specific and measurable, and DeepSeek to integrate technology-empowered inquiry processes. Collectively, they guide students to improve their instructional objective design capabilities through cycles of reflection and iteration.

Finally, there are two deep thinking sessions, shown in Fig. 4. In Session 1, students conduct bidirectional learning in-depth among themselves, and Session 2 involves in-depth reflection on educational research. Session 1 comprises outstanding and advanced students as double-ended learners who ask inspiring questions about the high level of difficulty arising from class Q&A and discussions. In Session 2, the teacher directs questions to develop the students' innovative research thinking and practice capabilities:" Is it feasible to design a procedural paradigm for the design of instructional objectives?"and "Can an indicator system be constructed to determine the corresponding levels of instructional objectives?"

## 5. Results and findings

Based on the three-dimensional framework of smart education, learning effectiveness, ICT, and classroom organization, combined with the educational requirements and characteristics of the M.Ed., we researched the characteristics of smart classes for M.Ed. programs. We found that, when setting educational goals, we should consider the three levels of knowledge content, research ability, and practical ability, and determined that the use of ICT should improve and realize the interaction, efficiency, and openness of the class. In a teaching organization, the class should have the characteristics of diversified learning forms, personalized learning content, cooperative learning elements, diversified learning assessment, and iterative cycles of discussion.

Through a comparative experiment and data analysis of teaching modes, we found that the smart class teaching mode had significantly better teaching effects than traditional class teaching. The smart teaching platform recorded all the data and class video recording,

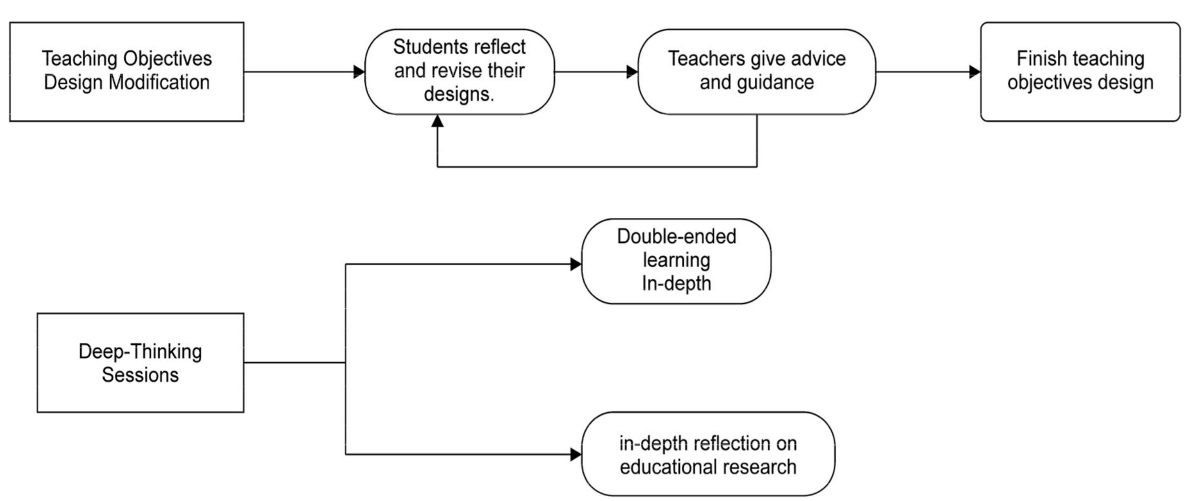


**Fig. 4.** After-class interaction model of smart teaching.

collected and organized the design of instructional objectives submitted by the two groups of students after class, and provided the teachers with the design results. Table A.5(see in the Appendix) shows the first-level index and total scores averaged across the three expert teachers' scores. Table A.6(see in the Appendix) shows the scores corresponding to each second-level index, and Fig. 5 shows the changes in the scores of the second-level indexes of the teaching-goal designs.

Through Table A.5, we can intuitively compare the differences between the traditional and smart class teaching methods. The total score of the experimental group corresponding to the smart class teaching mode was 4.06 points higher than that of the control group, which experienced the traditional teaching mode. Combined with Table A.6, we can see that the scores for each primary index in the experimental group were higher than those in the control group, and the scores corresponding to each secondary index were higher than those in the control group. Compared to the traditional teaching mode, the smart class significantly improved students' learning levels.

An independent samples *t*-test was conducted using IBM SPSS Statistics 26 to evaluate the differences in instructional objective design scores between the experimental and control groups, thereby comparing student learning outcomes under smart classroom versus traditional teaching models.As presented in Table A.7, the analysis revealed a statistically significant difference between the two groups. The experimental group ($M = 84.15$, $SD = 6.48$) outperformed the control group ($M = 80.09$, $SD = 6.47$), with results indicating $t\ (66) = 2.576$ and $p = 0.012$.Furthermore, the magnitude of this difference was substantial. The effect size was calculated as Cohen's $d = 0.62$, representing a medium-to-high effect level, with a 95% confidence interval of [0.12, 1.12].ConclusionThese findings demonstrate that student performance under the smart classroom model is superior to that under the traditional model. Consequently, it can be concluded that the smart classroom approach not only yields a statistically significant improvement in students' instructional objective design capabilities but also holds substantial practical educational significance.

An independent samples *t*-test was conducted on the scores for the three first-level indicators: curriculum standards, mathematics textbooks, and student learning conditions. As shown in Table A.8 (see in the Appendix), the probability values for the differences were all statistically significant ($p < 0.05$). These results indicate that, under the smart classroom model, students demonstrated significantly superior competency in instructional objective design across all three dimensions compared to those in the traditional teaching model.

Fig. 5 presents a comparison of scores between the experimental group and the control group across various secondary indicators, illustrating the specific effects of smart classroom teaching on enhancing students' instructional objective design proficiency. Among the secondary indicators, substantial differences in scores were observed in "competency development," "holistic integration," "content abstraction," "knowledge connotation," "knowledge foundation," and "thought experience." The significant advantage of the experimental group in 'T2 Competency Development' and 'T5 Knowledge Connotation' provides direct evidence of the deepening of their "professionalism" in instructional objective design. Meanwhile, the improvement in 'T1 Requirement Attainment' (+2.86) reflects progress in the "precision" of objective formulation. This indicates that the 'D-T-E Model,' through multidimensional technological support and multiple rounds of evaluation and reflection, has systematically fostered the synergistic development of students' core qualities of "precision" and "professionalism." The above analysis yields the following findings.

The D-T-E model's provision of open learning resources and

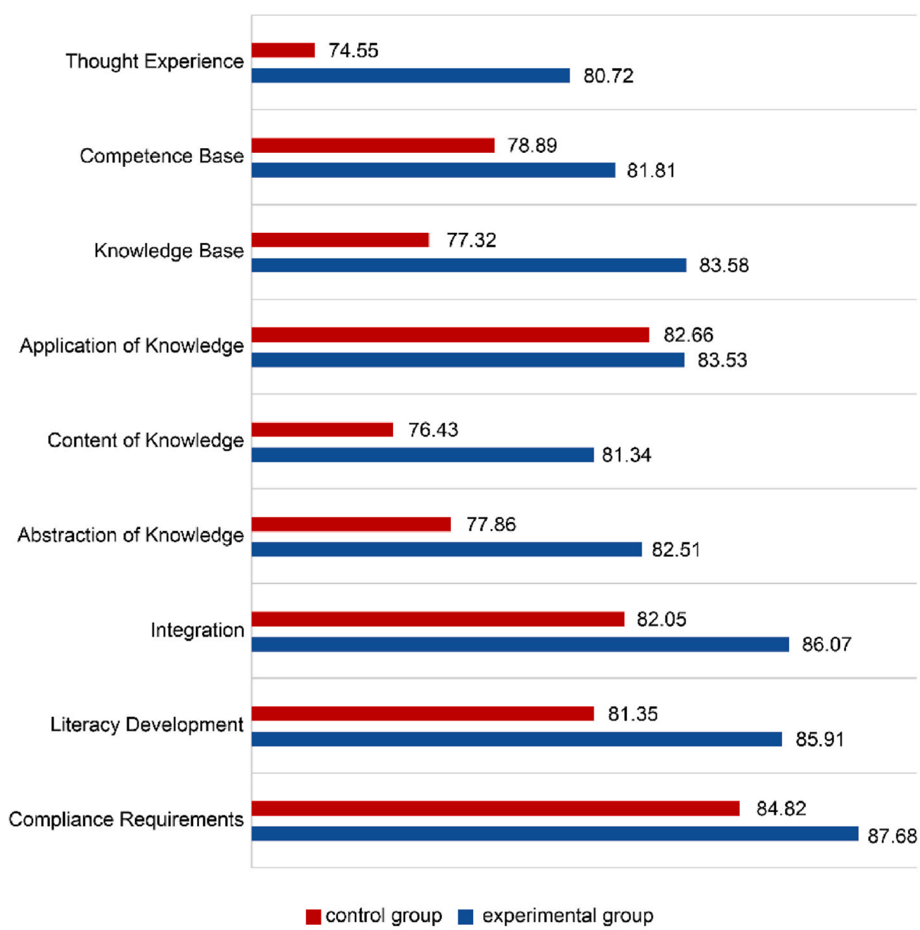


**Fig. 5.** Comparison of secondary indicator scores.

personalized content recommendation significantly deepened students' understanding of disciplinary core competencies, enabling them to formulate aligned instructional objectives accurately. Furthermore, the smart teaching platform effectively bridged information gaps by supplementing necessary materials, thereby helping students clearly identify and account for learners' prior knowledge foundations during the design process.

The diversified learning assessments and the iterative revision sessions after class, focused on instructional objective design, enable Master of Education students to continually refine deficiencies in their objective design and effectively integrate the instructional objectives across the three-dimensional goal framework. Consequently, the instructional objective design of the experimental group is generally superior to that of the control group.

The recursive interaction and multi-element collaboration within the "D-T-E Model" facilitated extensive communication among Master of Education students and between students and instructors. This process enables the sharing of perspectives on students' cognitive experiences, the abstraction process of mathematical content, and the essence of mathematical knowledge, thereby deepening the students' understanding of instructional objective design.

## 6. Discussion and conclusions

The value of the "D-T-E model" is realized through discipline-specific technological frameworks, a principle validated by our mathematics case study. Consistent with the ICT dimension of our framework, the experimental group utilized a platform tailored to the characteristics of mathematics—namely its "abstraction, logic, and applicability"—featuring functions such as a curriculum standards interpretation database. This group significantly outperformed the control group across all first-level indicators, most notably in the dimension of Student Learning Status (see Table A.5 in the Appendix). This finding supports the perspective of Tikhomirov et al. (2015) that ICT serves as a tool for achieving specific educational outcomes, standing in contrast to the generic technologies used by the control group. Furthermore, the experimental group's superior performance on second-level indicators such as "Content Abstraction" (see Table A.6) illustrateshow tailored tools can deepen understanding of mathematical conceptional experiences, aligning with the smart education attributes proposed by Yun et al. (2017). While the logic of matching disciplinary needs with technological functions is generalizable to other fields, it necessitates the adaptive modification of generic technologies (e.g., LLMs) to integrate core disciplinary features.

The enhancement of instructional effectiveness stems from a paradigm shift in evaluation based on the concept of "Assessment as Learning," which transforms evaluation into a tool for fostering metacognitive development—the core objective of the Learning Effectiveness dimension. The experimental group scored 4.56 and 4.02 points higher than the control group on the second-level indicators of "Competency Development" and "Integration & Alignment," respectively. Unlike the control group's summative, one-way feedback, the experimental group utilized the smart platform to create a multi-turn, interactive system encompassing intelligent diagnostics, peer assessment, and targeted instructor guidance. This established a “Design-Evaluation-Reflection-Revision”closed loop cultivating students' metacognitive skills and equipping them with methods for self-improvement (thereby validating Javaid et al., 2022; Glukhov & Vasetskaya, 2017, pp. 17–21).This shift also redefines "Learning Effectiveness," evidenced by the higher frequency of active learning behaviors observed in the experimental group, marking a transition from "passive reception of evaluation" to "active utilization of evaluation."

From a pedagogical perspective, the pre-class, in-class, and post-class process in this study adheres to the classic structure of "Preparation-Instruction-Practice-Feedback." Its underlying logic is not entirely disconnected from the traditional didactic teaching represented by the control group. The true breakthrough of this study lies in utilizing intelligent technologies to significantly augment the "feedback loop" within this classic structure. By providing immediate, multidimensional, and actionable feedback via AI, the study reinforced the iterative learning process of "Design-Evaluation-Reflection-Revision." This aligns closely with the "reflective practice" emphasized in constructivist learning theory and can be viewed as a deepening and extension of technology-enabled "Constructive Alignment."

Furthermore, this model restructures classroom interaction dynamics. The integration of the smart platform has extended teacher-student interaction from singular classroom Q&A to diversified, multi-turn exchanges based on multimedia resources, online forums, and collaborative tools. This technology-mediated interaction model creates a richer environment for students to deeply understand knowledge connotations and exchange ideas and experiences. In contrast to traditional teaching, which adopts a linear, closed, and didactic organizational model—leading to superficial understanding and limited integration of experiences, the experimental group constructed a multi-agent, multi-turn, and multi-media model based on the smart platform. In sessions such as "Knowledge Connotation Learning," "Idea & Experience Exchange," and "Collaborative Design," students deepened their learning through multimedia resources, cross-class forums, and online collaborative tools. Technology thus transforms into an interactive medium, constructing a communication environment with specific contextual characteristics and spatial-temporal flexibility, thereby enhancing the multi-dimensional thinking abilities of M.Ed. students.

This study has several limitations. Firstly, the research findings are primarily grounded in a case study of master's students in mathematics education. While this specific disciplinary context facilitates an in-depth and targeted inquiry, it may also constrain the direct generalizability of the results to other subject areas (such as the humanities or language disciplines) without further adaptation in terms of technology and instructional design. Secondly, regarding the research sample, two specific limitations exist. 1.Statistical Precision although the findings reached statistical significance ($p = 0.012$) with a medium-to-large effect size (Cohen's $d = 0.62$), the 95% confidence interval for Cohen's d ranged from 0.12 to 1.12, indicating considerable width. This imprecision is primarily attributed to the relatively small sample size ($N = 68$). Consequently, while these results affirm the positive impact of the intervention, caution is warranted in precisely estimating the magnitude of the effect.2.Sample Representativeness, the overall research sample presents two notable constraints: its relatively small size of 68 participants may not adequately capture the diversity within the broader population of mathematics M.Ed. students, and its strong regional and institutional concentration may limit the broader applicability of the findings across programs with varying training objectives, curricula, and instructional resources.

Looking forward, the continuous evolution of ICT promises to further transform classroom support and the very form of smart classrooms. Future work should prioritize the following directions: (1)Examining Transferability: Extending and adapting the proposed “D-TEL” model and smart teaching mode to other disciplinary and institutional contexts to verify their transferability; (2) Developing Assessment Frameworks: Constructing and validating a dedicated smart education evaluation framework that aligns with its core dimensions (learning effectiveness, ICT integration, and classroom organization); (3)Exploring Technological Frontiers: Investigating the potential of emerging technologies (e.g., generative AI, immersive environments) to dynamically refine the smart teaching system along the path of objectives, means, organization, modes, and evaluation; (4)Strengthening Empirical Validation: Expanding the sample size and conducting multi-center trials to narrow the confidence interval, thereby obtaining more robust and precise effect estimates, so as to better validate the effectiveness and generalizability of this model.

## CRediT authorship contribution statement

**Fusheng Zhu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Qian liang:** Validation, Software, Investigation. **Zhanmin Mao:** Software, Funding acquisition. **Ying Wang:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition.

## Availability of data and materials

Not applicable.

## Declaration of ethical approval

The study was approved by an ethical committee with ID: WHU-HSS-IRB2026026. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by sending request e-mails to the corresponding author.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Doubao in order to improve language, readability and organization/flow of this article. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

Not applicable.

## Appendix A

**Table A.1**
Evaluation index system for the design of instructional objectives in high school mathematics

| Level 1 indicators | Secondary indicators | Evaluation criteria | Weights |
|---|---|---|---|
| Curriculum standard | T1 Compliance requirements | The instructional objectives are designed to meet the requirements of the new Curriculum Standard for the mastery of relevant mathematical content, clearly state the level that students are required to achieve, reflect the hierarchical nature of the objectives, and describe them with easy-to-measure behavioral verbs. | 0.180 |
| | T2 Literacy development | The instructional objectives are formulated to highlight the core literacy of the Mathematics discipline, which can be correctly expressed as the development of students' quality of thinking, essential character, key mathematical competencies, and affective attitudes and values related to the content of knowledge through specific activities, to enable the students to gradually form correct values and develop core literacy in the mathematics discipline in the process of learning and applying mathematics. | 0.139 |
| | T3 Integration | The design of the instructional objectives reflects the organic integration of the three goals based on the mastery of specific concepts and the formation of corresponding skills, integrating the relevant ideas and methods, activities, and experience, as well as the requirements of the core literacy of the discipline of mathematics, reasonably describing the connection between the concepts, and forming a reasonable level of the content system. | 0.102 |
| Mathematics textbooks | T4 Abstraction of knowledge | The instructional objectives reflect that teachers design appropriate problem situations, provide students with opportunities for independent exploration and sufficient space for reflection, and allow students to experience the formation and development of knowledge through observation, manipulation, experimentation, generalization, and analysis. | 0.103 |
| | T5 Content of knowledge | instructional objectives reflect the essential attributes, connotation and extension, and logical structure of mathematical knowledge and explicitly express the cultivation of students to grasp the content of mathematical knowledge. | 0.150 |
| | T6 Application of knowledge | The instructional objectives are designed to reflect the specific application areas and the application level of mathematical concepts in senior secondary school. The selection of applications is in line with the age characteristics of students and is in close touch with the reality of production and life at present. | 0.104 |
| Student learning | T7 Knowledge base | The instructional objectives are designed to fit high school students' mathematical knowledge base to learn mathematics. | 0.101 |
| | T8 Competence base | The instructional objectives are designed to fit the foundation of mathematical abstraction, induction, reasoning, and other mathematical skills that high school students have to learn mathematics. | 0.070 |
| | T9 Thought experience | The instructional objectives are designed to fit the basic ideological foundations of combining numbers and shapes, transforming them, and delimiting boundaries that high school students possess when learning mathematics, as well as the basic experiential foundations of activities such as independent thinking, independent inquiry, cooperative communication, discovering and posing problems, and analyzing and solving problems. | 0.051 |

**Table A.2**
Evaluation Scale for Designing High School Mathematics instructional objectives

| Evaluation dimension | Evaluation index | Scoring criteria |
|---|---|---|
| 1. Scientificity of objectives (30%) | 1.1 Align with curriculum standards | It is consistent with the content of the relevant modules in the "General High School Mathematics Curriculum Standards (2017 Edition)" |
| | 1.2 The discipline is logically rigorous | Pay attention to the internal logic and interconnections among concepts, theorems, and formulas |
| | 1.3 No scientific errors | Emphasis on mathematical knowledge and precise use of mathematical language expression |
| 2. Target adaptability (25%) | 2.1 Align with students' cognitive level | Alignment with the class's specific context and current learning level of the class |
| | 2.2 Adapt teaching content | Focus on key knowledge points and core mathematical thinking methods |
| | 2.3 Take individual differences into account | Fully consider the diverse needs of different students |
| 3. Operability of the goal (25%) | 3.1 Express clearly and specifically | The use of action verbs is precise (such as demonstrate, apply, prove, solve, analyze, etc.) |
| | 3.2 Detectable and evaluable | Clarify the types of exercises, task formats, and other assessment modalities |
| | 3.3 Adaptive teaching process | The teaching process is complete, and the teaching approach is clear |
| 4. The developmental nature of the goal (20%) | 4.1 Core competency orientation | A student-centered approach that fully integrates core competencies |
| | 4.2 Promote ability improvement | Integration of key cognitive demands: logical reasoning, procedural fluency, and problem solving |
| | 4.3 Connecting with subsequent learning | Pay attention to the vertical extension and horizontal correlation of knowledge |

**Table A.3**
Class activities in designing high school mathematics instructional objectives.

| Session | Concern | Student Activities | Teacher Activities | Smart Teaching Platform |
|---|---|---|---|---|
| Session 1 | What do you think is the basis for the instructional objectives set in the curriculum standards? | Answering voluntarily and sharing insights | Responding to students' answers | Recording the class |
| Session 2 | Do you have questions about the setting of instructional objectives in the curriculum standards? | Inter-group discussions | Observing students' discussions and helping appropriately | Monitoring students' discussions and conducting analysis |
| Session 3 | Filtering and organizing from the instructional objectives given by the teacher and setting a time frame for the completion of each objective in the course | Completing the task themselves or collaborating between groups | Observing students' task completion and analyzing the data results provided by the Smart Teaching Platform | Providing a task submission platform, counting and analyzing the tasks submitted by students, and giving teachers and students feedback |
| Session 4 | Discussing the differentiation in setting time for completion of instructional objectives | Thinking and discussing | Organizing a discussion among students | Recording the class |

**Table A.4**
Comparison table of three AI models: DeepSeek, Kimi, and Doubao

| AI model | DeepSeek | Kimi | Doubao |
|---|---|---|---|
| Style analysis | Academic rigor | Emphasis on details | Concise and efficient |
| Core perspective | From the perspective of curriculum theory: focusing on the structure, logic, and normativity of the objectives. | From the perspective of teaching methodology: focusing on students' cognition and practical feasibility behind the objectives. | From the perspective of evaluation: providing a directly measurable and evaluable solution. |
| Comment on the homework designed for instructional objectives | 1 Focus on core knowledge, clarify key skills, and adhere closely to the "definition expression" and "mathematical methods".<br>2 Standardize action verbs, ensure that behavioral expressions are observable and measurable, and meet the requirements of operability.<br>3 The learning path is scientific, following the cognitive law of moving from the concrete to the abstract and from intuition to logic.<br>4 Balance mathematical aesthetics and application value with a high-level goal. | 1 Analyzing from four dimensions: knowledge type, cognitive level, literacy orientation, and measurability, instructional objectives encompass conceptual knowledge, procedural knowledge, strategic knowledge and literacy orientation.<br>2 The cognitive levels are categorized as understanding, application, comprehending, and synthesis.<br>3 The first three objectives are measurable, while the fourth objective is not. | 1 Emphasize the core knowledge of function monotonicity, clearly define expressions, and highlight key skills such as using graphs to judge monotonicity.<br>2 Follow the cognitive rule of moving from intuition to abstraction, use examples and images to aid conceptual understanding, strengthen the use of symbolic language.<br>3 Balance the improvement of core competencies and the perception of mathematical culture, pay attention to the simplicity of symbols, and reflect a competency-oriented approach. |
| Suggestions for revising the homework designed based on instructional objectives | 1 Accurately express terms, and clarify the symbolic function of mathematical language.<br>2 Refine the scope of objectives, and specify the monotonicity of function on a specified domain.<br>3 Strengthen process design, and guild students to master skill - "how to do it".<br>4 Visualize literacy objectives, and clarify the specific implementation direction of | 1 Clarify the expression form of symbolic language.<br>2 Distinguish between strict monotonicity and non-strict monotonicity.<br>3 Add observable behavioral descriptions. | 1 Clearly define "mathematical language" to the point of monotonicity, supplement the requirements for determining the monotonicity of linear and quadratic functions, and enhance precision and operability.<br>2 Refine instance types and clarify specific training methods for symbolic language characterization.<br>3 Combine activities such as independently designing solutions to practical problems to |

(continued on next page)

**Table A.4** (*continued*)

| AI model | DeepSeek | Kimi | Doubao |
|---|---|---|---|
| | core competencies in conjunction with the content of this lesson. | | strengthen the connection between mathematics and realities. |
| Model advantages | The framework is complete, the analysis is comprehensive, the theoretical foundation is strong, and the provided version adheres to standards and norms. | Strong insight, directly pointing to the core issues. | Focusing on the "measurability" of the goal, the output results are clear and direct. |
| Model drawbacks | The analysis from the students' perspective lacks depth. | The analysis process is skipped, and the language fluency is weak | The analysis process is simplified, lacking elaboration on the logic of optimization. |

**Table A.5**
Tier 1 indicator scores and total scores

| | Scores in Curriculum Standard | | Scores in Mathematics Textbook | | Scores in Student Learning | | Overall Scores | |
|---|---|---|---|---|---|---|---|---|
| | Experimental group | Control group | Experimental group | Control group | Experimental group | Control group | Experimental group | Control group |
| Average | 86.60 | 82.82 | 82.34 | 78.78 | 82.38 | 77.00 | 84.15 | 80.09 |

**Table A.6**
Average scores of secondary indicators

| | Scores in Curriculum Standard | | | Scores in Mathematics Textbook | | | Scores in Student Learning | | |
|---|---|---|---|---|---|---|---|---|---|
| | Compliance Requirements | Literacy Development | Integration | Abstraction of Knowledge | Content of Knowledge | Application of Knowledge | Knowledge Base | Competence Base | Thought Experience |
| Experimental group | 87.68 | 85.91 | 86.07 | 82.51 | 81.34 | 83.53 | 83.58 | 81.81 | 80.72 |
| Control group | 84.82 | 81.35 | 82.05 | 77.86 | 76.43 | 82.66 | 77.32 | 78.89 | 74.55 |

**Table A.7**
Independent-samplest-test for total scores.

| Groups | Average | t | Sig. |
|---|---|---|---|
| Experimental group | 84.15 | 2.576 | 0.012 |
| Control group | 80.09 | | |

**Table A.8**
Independent-samplest-test for Tier 1 indicator scores.

| Level 1 indicators | Groups | Average | t | Sig. |
|---|---|---|---|---|
| Curriculum Standard | Experimental group | 86.60 | 2.351 | 0.022 |
| | Control group | 82.82 | | |
| Mathematics Textbooks | Experimental group | 82.34 | 2.111 | 0.039 |
| | Control group | 78.78 | | |
| Student Learning | Experimental group | 82.38 | 3.434 | 0.001 |
| | Control group | 77.00 | | |

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